# Angstrong Nuwa-HP60C

**Angstrong Nuwa-HP60C**

1. Content Description
2. Camera Introduction
3. Preparation
4. Case Demonstration
    4.1 View Camera Node Information
    4.2 Subscribe to the Color Image
    4.3 Subscribe to the Depth Image
    4.4 Visualize the Point Cloud

## 1. Content Description

> **Important**
>
> - This section is for reference only for users of the Angstrong Nuwa-HP60C camera

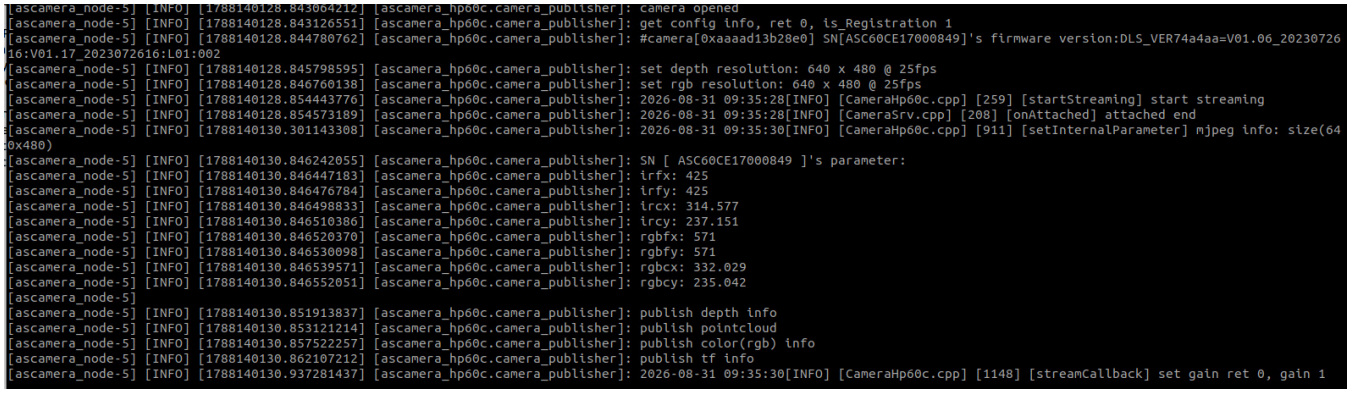
- This tutorial introduces the Angstrong Nuwa-HP60C used on the product, and how to view the color image, depth image, infrared ir image, and point cloud published by the camera node.

## 2. Camera Introduction



## 3. Preparation

```
ros2 launch yahboomcar_bringup hp60c_view.launch.py
```




# 4. Case Demonstration

## 4.1 View Camera Node Information

You can use the `ros2 node` tool to view which topics the camera node publishes and which services it provides. Enter in the terminal,

```
ros2 node info /ascamera_hp60c/camera_publisher
```

The published topics are shown in the figure below

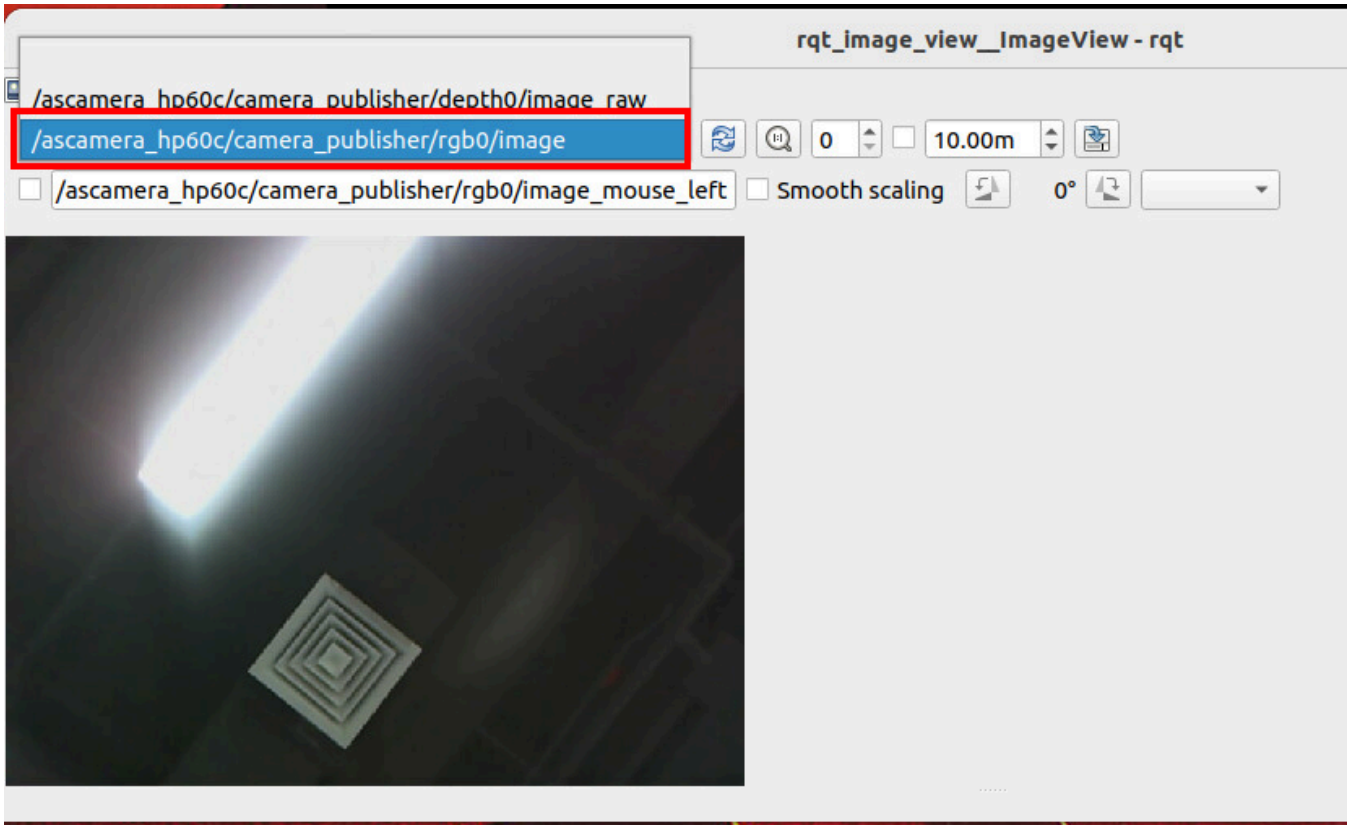
```
/ascamera_hp60c/camera_publisher
  Subscribers:
    /parameter_events: rcl_interfaces/msg/ParameterEvent
  Publishers:
    /ascamera_hp60c/camera_publisher/depth0/camera_info: sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/depth0/image_raw: sensor_msgs/msg/Image
    /ascamera_hp60c/camera_publisher/depth0/points: sensor_msgs/msg/PointCloud2
    /ascamera_hp60c/camera_publisher/rgb0/camera_info: sensor_msgs/msg/CameraInfo
    /ascamera_hp60c/camera_publisher/rgb0/image: sensor_msgs/msg/Image
    /parameter_events: rcl_interfaces/msg/ParameterEvent
    /rosout: rcl_interfaces/msg/Log
    /tf: tf2_msgs/msg/TFMessage
```

The provided services are shown in the figure below,

```
  Service Servers:
    /ascamera_hp60c/camera_publisher/describe_parameters:
rcl_interfaces/srv/DescribeParameters
    /ascamera_hp60c/camera_publisher/get_parameter_types:
rcl_interfaces/srv/GetParameterTypes
    /ascamera_hp60c/camera_publisher/get_parameters: rcl_interfaces/srv/GetParameters
    /ascamera_hp60c/camera_publisher/list_parameters: rcl_interfaces/srv/ListParameters
    /ascamera_hp60c/camera_publisher/set_parameters: rcl_interfaces/srv/SetParameters
    /ascamera_hp60c/camera_publisher/set_parameters_atomically:
rcl_interfaces/srv/SetParametersAtomically
```

## 4.2 Subscribe to the Color Image

- After the program starts, it will automatically open **rqt_image_view** and display the image




## 4.3 Subscribe to the Depth Image

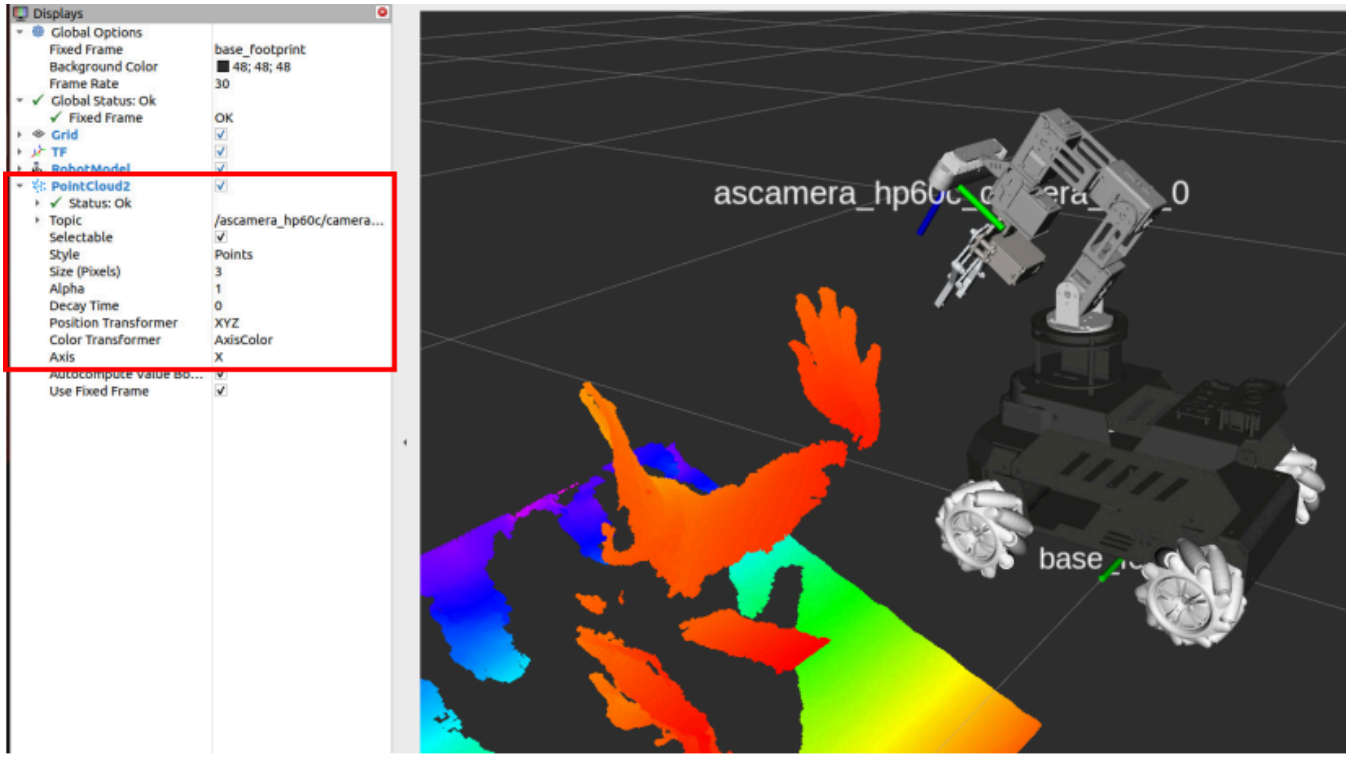
- In **rqt_image_view**, select the depth image topic to display the depth image




As shown in the figure above, this displays the depth image of the `/ascamera_hp60c/camera_publisher/depth0/image_raw` depth image topic.

## 4.4 Visualize the Point Cloud

- In the rviz window you can observe the robot car model and the corresponding environment point cloud, and you can modify the shape and color of the displayed point cloud by changing style and Axis


Displays
Global Options
Fixed Frame
base_footprint
Background Color
48; 48; 48
Frame Rate
30
Global Status: Ok
Fixed Frame
OK
Grid
TF
RobotModel
PointCloud2
Status: Ok
Topic
/ascamera_hp60c/camera...
Selectable
Style
Points
Size (Pixels)
3
Alpha
1
Decay Time
0
Position Transformer
XYZ
Color Transformer
AxisColor
Axis
X
Use Fixed Frame